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When AI is wrong: the limits of human oversight in AI-assisted diagnostic decision-making

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ABSTRACT

The integration of artificial intelligence (AI) assistance in the workplace raises concerns about its reliability in complex decision-making tasks. While AI systems can make mistakes, human oversight, particularly human gatekeeping correction, is often proposed as a solution to mitigate errors. However, it remains unclear whether humans can effectively identify and correct AI errors, especially in critical and nuanced decisions like medical decision-making. In this study, we investigate the effects of AI assistance accuracy on diagnostic work performance using Traditional Chinese Medicine (TCM) as an example. Our results show that incorrect AI outputs harm doctors' decision-making and reduce diagnostic accuracy. Additionally, we explore the role of human factors, such as engagement in long logical thinking, resistance to AI, and professional expertise. These findings challenge the feasibility of relying on human oversight and provide new insights into the practical implications of AI-assisted decision-making in the workplace.

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Human oversight; AI accuracy; human-AI collaboration; decision-making; Traditional Chinese medicine

1. Introduction

Recently, the employment of artificial intelligence (henceforth, AI) assistive systems in workplace has emerged as a transformative force across industries (Murray, Rhymer, and Sirmon 2021). It denotes a seminal technique substitution (Yang, Ngai, and Wang 2024). AI assistance has demonstrated significant potential to enhance human work performance as a decision-augmenting tool, for example, in the healthcare industry (e.g. Perez et al. 2024; Lebovitz, Lifshitz-Assaf, and Levina 2022; Patel et al. 2019; Abdulrahman, Richards, and Bilgin 2023; Jin and Eastin 2025), creative production (e.g. Magni, Park, and Chao 2024; Van Esch, Black, and Ferolie 2019) and consulting services (e.g. Dell'Acqua et al. Forthcoming).

As AI assistance becomes increasingly integrated into complex decision-making tasks, concerns about its reliability and potential consequences are growing, particularly in fields requiring nuanced judgments such as medicine (Triantafyllopoulos et al. 2024). AI reliability can be assessed through several factors, including

accuracy, explainability, ethical fairness, and user confidence (Ma et al. 2023). Among these, accuracy is a fundamental measure of AI reliability (Rechkemmer and Yin 2022; Balagurunathan, Mitchell, and El Naqa 2021). However, AI accuracy remains vulnerable to issues like imperfect algorithms and limited data (Izadi and Forouzanfar 2024). Kim, Oh, and Lee (2024) also point to societal concerns about biased AI and automated decision-making.

Given that AI outputs cannot be directly manipulated, errors in AI-generated content pose significant risks in the workplace decision-making (e.g. Johnson et al. 2023; Cappellani et al. 2024). This highlights the critical role of human oversight in AI-assisted decision-making, where practitioners must assess AI recommendations to mitigate potential errors, fostering effective human-AI collaboration (Magni, Park, and Chao 2024; Choudhary et al. 2025). Recent scholarship has raised fundamental questions about the feasibility of human oversight in AI systems, highlighting structural and cognitive limits that may render effective

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supervision impossible in certain contexts (Holzinger et al. 2025a). While Holzinger et al. raise a broader conceptual concern about whether human oversight remains feasible, our study provides experimental evidence on the conditions under which such oversight breaks down in AI-assisted diagnosis. This addition strengthens the paper's theoretical positioning within the literature on human–AI collaboration and the limits of oversight.

While previous studies have hinted at shifts in reasoning patterns (Jussupow et al. 2021) and raised concerns among medical practitioners when AI errors influence diagnostic decisions (Abdel-Karim et al. 2023), they have not established a direct causal link between AI accuracy and decision-making work performance. Therefore, it is essential to examine how well humans can evaluate AI advice and respond to false outputs in decision-making contexts. This paper aims to address this gap in the literature.

To address these questions, we examine AI-assisted diagnostic decision-making in Traditional Chinese Medicine (TCM). TCM diagnosis exemplifies a complex and nuanced form of decision-making that relies heavily on dialectical reasoning and experiential knowledge (Zhang, Zhou, and Zhang 2021; Tang, Liu, and Ma 2008). Unlike Western medicine, TCM is often described as an experience-based medicine, where doctors' subjective interpretations play a critical role in forming diagnoses. This context provides a unique lens through which to investigate the effects of human oversight when false AI advice is introduced into human–AI collaboration. Therefore, we conducted an experiment evaluating TCM doctors' diagnostic work performance and explored how it changes when imperfect AI assistance (i.e. incorrect advice) is provided. Furthermore, given the substantial heterogeneity in how AI assistance affects individual work performance (Brynjolfsson, Li, and Raymond 2025), we also investigate additional factors that may moderate this relationship. Accordingly, our study seeks to answer two research questions: (1) *What is the effect of human oversight on imperfect AI-assisted TCM diagnosis* and (2) *What moderating factors influence the relationship between AI assistance accuracy and diagnostic work performance?*

The contribution of this research is threefold. First, this study addresses a critical research gap concerning human oversight in the human–AI collaboration literature by investigating TCM diagnosis, an example of complex and nuanced decision-making task. We offer novel insights from the underexplored effects of human oversight in AI-assisted diagnostic decision-making. Second, our study explores the boundary conditions of the causal relationship between AI assistance's

accuracy and TCM diagnostic work performance. We test and discuss the latent moderated roles of human factors, and we pay special attention to how human responses to incorrect AI advice and whether people can serve as effective gatekeepers for AI assistance in the workplace. Finally, we provide empirical evidence that practitioners ought to take a more prudent attitude towards AI assistance, highlighting the risks of causing cognitive bias and over-reliance on imperfect AI algorithms in decision-making tasks. Our findings can foster a better use of advanced AI assistance in work and enhance human–AI collaboration across industries.

2. Theoretical background and hypotheses

2.1. AI assistance and human oversight

AI assistance refers to the integration of AI techniques like machine learning algorithms and natural language processing into workplace systems (Brynjolfsson, Li, and Raymond 2025). Such AI assistive tools range from chatbots to generative AI models like ChatGPT. The literature identifies two primary roles for AI assistance: automation and augmentation (Raisch and Krakowski 2021). Automation implies the substitute of AI algorithms for human workers, whereas augmentation entails enhancing human work performance through AI support. Current research claims that AI has not yet achieved the level of stand-alone performance required for prediction-based managerial decision-making; thus, AI is predominantly used as a decision-augmenting tool in complex decision contexts (Choudhary et al. 2025). For instance, the 'augmented radiologist' framework explicitly assumes that expert clinicians can contextualise, verify, and correct AI outputs (Sorantin et al. 2022). Former empirical studies have proved the capacity of AI assistance (after training and model fine-tuning) in healthcare industry to support and improve radiological diagnosis by providing diagnostic advice or visual explanations (Perez et al. 2024; Lebovitz, Lifshitz-Assaf, and Levina 2022; Patel et al. 2019). However, this assumption relies on the premise that human experts reliably function as effective 'gatekeepers' when AI advice is incorrect. Our study does not reject the augmentation model per se; rather, we interrogate this core assumption.

Nonetheless, despite advancements, AI assistance is not infallible. Accuracy is a critical attribute of AI reliability (Rechkemmer and Yin 2022; Balagurunathan, Mitchell, and El Naqa 2021). It depends on algorithmic design, data quality, contextual applicability, among other factors (Izadi and Forouzanfar 2024). Challenges such as algorithmic opacity, biased training data and

overfitting can lead to errors of AI outputs. For instance, previous studies have concluded that AI chatbots such as ChatGPT sometimes provide incorrect, incomplete and potentially harmful information about common medical conditions, indicating its important limitations in medical queries (Johnson et al. 2023; Cappellani et al. 2024). Moreover, AI remains limited in its ability to replicate core human attributes such as ethical reasoning, contextual judgment, and emotional intelligence. However, professions requiring nuanced decision-making, such as management, healthcare, law, and venture capital, rely on human expertise to interpret ambiguous scenarios, weigh moral trade-offs, and build trust (Budhwar et al. 2023). For example, a physician's diagnosis integrates not only clinical data but also patient history, cultural context, and empathy, which are elements beyond AI's algorithmic reach (Nadarzynski et al. 2019). Therefore, Choudhary et al. (2025) propose a way of human-AI collaboration to optimise decision-making tasks: human oversight, where humans are responsible to check and rectify the outcomes from AI assistance to mitigate latent prediction errors; and humans are the ones to make the final decision. Overall, this approach effectively balances the unique strengths of both human judgment and AI ability.

2.2. AI-Assistive TCM diagnostic decision-making

TCM has been developed by Asian people for hundreds of years, and is now increasingly acknowledged and adopted around the world (Cheung 2011). In TCM diagnosis, there are four non-invasive methods that used to assess the physical conditions of patient: inspection (looking), auscultation (listening and smelling), interrogation (asking) and palpation (touching) (Zhang, Zhou, and Zhang 2021). TCM doctors usually conduct their diagnostic decision-making through the observations of the patient's visual characteristics and symptom inquiry, which is an unmeasurable and highly personalised process (Tang, Liu, and Ma 2008; Zhang, Zhou, and Zhang 2021). Moreover, in TCM diagnosis, the relationship between syndromes and diseases is dialectic: diseases can be linked to many syndromes, and vice versa (Zhang et al. 2020). Therefore, it requires TCM doctors to engage in substantial nuanced decision-making in diagnostic practices.

Recent advances in AI have been introduced to assist with medical diagnostic practices, including areas of health data collection and disease analysis (Bitkina, Park, and Kim 2023). Currently, AI assistance provides an opportunity for TCM doctors to improve diagnostic work performance by augmenting the precision of the data collection from traditional four diagnostic methods

and human decision-making (Wang et al. 2021; for reviews, see Zhang, Zhou, and Zhang 2021 and Zhang et al. 2023). For the improvement of decision-making itself, prior studies majorly employed natural language processing to provide doctors with advice of predicting disease and syndromes (Zhang et al. 2020). There are also empirical evidence from Western medicine showing that, when assisted with AI, the diagnosis accuracy of human doctors is significantly at a higher level (Wang et al. 2024; Patel et al. 2019; Cabitza et al. 2023).

However, previous studies implied that the effects of AI assistance as a decision tool in medical content can be varied. Prior study on Greek doctors reported only moderate satisfaction with ChatGPT's accuracy (Triantafyllopoulos et al. 2024). Jussupow et al. (2021) describe the diagnostic decision-making process of radiologists and document dynamic shifts between systematic and heuristic reasoning when physicians' judgments conflict with AI advice. Therefore, we expect that AI assistance's accuracy will cause significant effects in TCM diagnostic practices.

On the one hand, decision-makers like medical practitioners keep accumulating information until there is enough information to reach threshold to formulate a decision, leading to a Bayesian-like behaviour (Brocas 2012). In addition, there are empirical evidence reporting a Bayesian-like behaviour of endoscopists to follow the AI advice more when it is correct than incorrect (Reverberi et al. 2022). Thus, if provided with correct AI advice, diagnostic decisions will be augmented by TCM doctors naturally. On the other hand, however, Kahneman's study (2011) claims that human being is usually a cognitive miser with cognitive bias. Since TCM diagnosis is a complicated dialectic methodology per se (Zhang et al. 2020), once given conflicting views, people are vulnerable to cognitive bias, causing diagnostic errors. In this case, humans are likely to be influenced by false AI assistance and fail to remedy AI's mistakes. In other words, human oversight would lose efficacy. Therefore, we posit that the accuracy of AI assistive tool will cause significant negative effects on TCM doctor's diagnostic work performance, signalling the failure of human oversight:

Hypothesis 1: Incorrect AI advice harms TCM doctor's diagnostic work performance.

2.3. Influence of human factors

To further explore the effects on TCM diagnostic work performance, we should consider the influence of human factors. The dual-process theory is widely accepted as the dominant paradigm for understanding

medical decision-making (Croskerry 2009; Evans 2008). There is a consensus about the two different modes of thinking in dual-process theory: rapid intuitive thinking and long logical thinking (Kahneman 2011; see Evans 2008 for a review). Moreover, a functional near-infrared spectroscopy study observed the activation of clinical practitioners' prefrontal cortex when performing diagnoses, which indicates that long logical thinking is a cognitive process physiologically distinct from rapid intuitive thinking in medical diagnostic practices (Rotgans et al. 2019). As an experience-based medicine, TCM diagnosis relies largely on empirical methods and expertise (Liu et al. 2011). With the accumulation of clinical experience, expertise and personal practices, TCM doctors will acquire ample 'illness scripts', transforming long logical thinking to rapid intuitive thinking (Custers, Boshuizen, and Schmidt 1996). So, when performing diagnosis, TCM doctors activate the matching patterns of 'illness scripts' automatically, which is fast and intuitive; furthermore, it is understandable that senior TCM doctors should outperform in such matching practices (Wang and Yao 2009). Despite its speed, there is possibility that TCM doctors may miss information or fall into inertial thinking due to cognitive bias (Kahneman 2011).

Current AI assistance can provide a predictive diagnostic judgment of diseases and syndromes for TCM doctor's further analysis and consideration. With this help, we expect that TCM doctors will switch to long logical thinking more frequently during diagnostic decision-making. In this way, it is likely that TCM doctors are able to circumvent their own cognitive biases and be aware of diagnostic errors from incorrect AI advice. If TCM doctors recognise the mistakes made by AI assistance, they will rectify them and provide a correct diagnostic judgment. Therefore, we posit that TCM doctor's performing long logical thinking will influence the relationship between AI assistance's accuracy and diagnostic work performance:

Hypothesis 2: TCM doctor's performing long logical thinking moderates the causality between incorrect AI advice and poor diagnostic work performance, such that the relationship is weaker when the depth of long logical thinking is at higher level.

In addition, resistance refers to one's unfavourable response or rejection to change associated with a new information system technique, which is a pivotal factor that influences the application and spread of the technique (Markus 1983; Hirschheim and Newman 1988). The study of Talke and Heidenreich (2014) indicates that a deliberate innovation resistance evolves from

two innovation-specific factors: functional barriers and psychological barriers. Functional barriers arise when user perceives the attributes of innovative product as dysfunctional or inadequate for personal needs and usage expectations, while psychological barriers occurs as soon as the innovation conflicts with user's social norms, values and so forth. This measure is well-reexamined to identify resistance to AI in the current context (Yang, Ngai, and Wang 2024). Such perceived feelings to AI algorithms are solid even when individuals are informed of the AI errors (You, Yang, and Li 2022).

For individuals, when employing AI as a supportive tool for decision-making, different people may take different attitudes and actions on implementing AI advice (Duan, Edwards, and Dwivedi 2019). Moreover, existing literature shows that doctors' varying personal traits and understandings towards techniques result in different professional decisions (Borges and Gibson 2005). Thus, doctor's resistance to AI is predicted to influence the adoption of AI assistive advice, we then postulate that the level of TCM doctor's resistance to AI will negatively affect the relationship between the correctness of AI advice and diagnostic work performance:

Hypothesis 3: TCM doctors' resistance to AI moderates the causality between incorrect AI advice and poor diagnostic work performance, such that the relationship is weaker when resistance to AI is higher.

Furthermore, the experience level of practitioners is an important factor that influences the use of AI assistance (Cabitza et al. 2023). In a study of medical coding industry, senior workers had a lower productivity lift from AI assistance because of the distrust triggered by broader job responsibilities than junior ones (Wang, Gao, and Agarwal 2023). In another paper (Wang, Rau, and Yuan 2023) point out that **AI literacy / user competence** as a key individual factor. However, for senior TCM doctors, it is likely that they have difficulty in getting started with AI assistance and stick with their own diagnostic decision-making. In other words, the potential AI errors will have little impact on their work. Although former studies have documented the possible negative impact of work experience on the utilisation of new technologies (Tan and Netessine 2020), senior TCM doctors may avoid the drawbacks from such advanced techniques as well. Thus, we expect the work experience of TCM doctors will influence the relationship between AI assistance's accuracy and diagnostic work performance:

Hypothesis 4: Professional expertise moderates the effect of AI advice accuracy on diagnostic work performance, such that the detrimental effect of incorrect AI advice is weaker, and the beneficial effect of correct AI advice is stronger, among more professionally qualified participants.

Our conceptual framework is illustrated in Figure 1.

3. Method

We employed a randomised controlled experiment to investigate the causal effects of AI assistance accuracy on TCM diagnostic performance. Participants interacted with an AI chatbot via a custom-built web platform that enabled the systematic manipulation of advice correctness while maintaining interface consistency. Complementing the task performance data, we administered integrated surveys to capture participants' judgment confidence, depth of logical reasoning, resistance to AI, and demographic characteristics. These data were subsequently analysed using regression models to test our hypotheses.

3.1. Research design

3.1.1. Participants

We invited TCM doctors from cooperative hospitals in Guangdong, China, and recruited students from a medical school in Zhejiang, China. Eligible participants must have received standard TCM training. Moreover, we also administered a quiz of TCM knowledge before the experiment to check the qualifications of our participants. All of them passed the quiz. Eventually, we recruited a total of 49 TCM doctors and 73 medical students; however, seven of them withdrew halfway and five subjects submitted multiple evaluations of the cases in different groups, resulting in a total of 110 effective participants and 440 observations for analysis. Participants received compensation for their participation.

3.1.2. Procedure

The experiment comprised five sequential sections: (1) a qualification quiz, (2) AI prompts training, (3) Task 1, (4) Task 2, and (5) a post-experiment survey (see flowchart in Figure 2).

3.1.2.1. Preparatory sections. The first section was a qualification quiz assessing participants' knowledge of basic TCM diagnostic principles and skills. Only those who passed the quiz were considered eligible and proceeded to the subsequent sections.

The second section was an AI prompts training module (see Figure 3). This training was designed to help participants formulate effective queries, thereby promoting better human–AI interaction during the main experimental tasks.

After completing the preparations, participants entered the main part of the experiment, which consisted of Task 1 and Task 2. Both tasks required participants to diagnose clinical cases. To simulate real-world clinical reasoning, participants were instructed to provide a disease judgment and a corresponding prescription based on their own analysis (see Figure 4).

Throughout both tasks, participants could interact freely with an assistive AI chatbot (see Figure 5). The interface provided a suggested prompt: 'From the perspective of Traditional Chinese Medicine (TCM), please provide syndrome differentiation, pattern analysis, formula recommendations, and herbal/medication analysis'. AI advice was generated only when participants proactively submitted their queries to the chatbot.

All cases were selected from the National Qualification Examination for Medical Practitioners administered by the National Administration of Traditional Chinese Medicine and were matched for difficulty level. Although exam-style cases rely on written vignettes rather than live patient interaction, they comprehensively present clinical data derived from all four dimensions of TCM diagnosis (inspection, auscultation, interrogation, and palpation). Thus, solving these cases requires the same core clinical reasoning workflow used in real practice: synthesising multi-modal diagnostic information to form a judgment.

3.1.2.2. Task 1: real AI interaction (ERNIE). Task 1 consisted of two cases (Case 1 and Case 2). In this task, we employed the ERNIE-3.5-8 K algorithm (Baidu), a general-purpose Large Language Model (LLM) that could not directly search the internet for the experiment case answers. Participants were randomly assigned to either the Control group (no AI assistance) or the ERNIE group. Across the two cases, each participant experienced both conditions exactly once, with the order counterbalanced: half of the participants completed Case 1 with their assigned condition followed by Case 2 with the alternative condition, while the other half completed the reverse sequence.

3.1.2.3. Task 2: manipulated AI advice (Correct vs. Incorrect). Task 2 also consisted of two cases (Case 3 and Case 4). In contrast to Task 1, the presentation

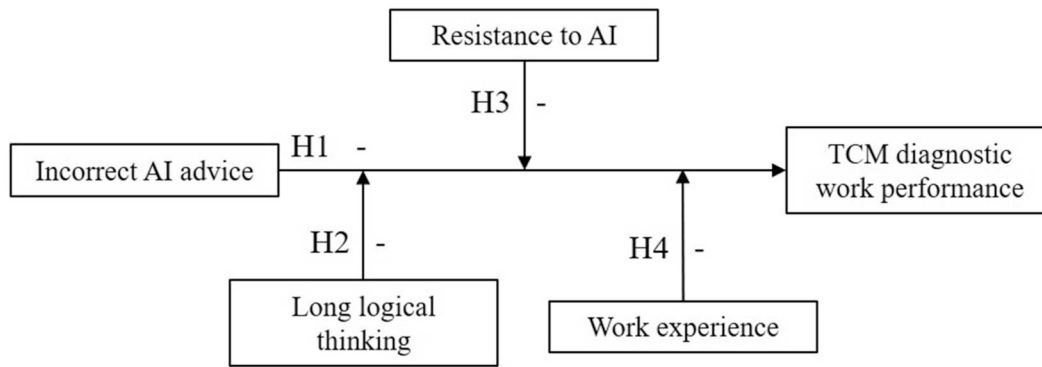


Figure 1. Conceptual framework.

order was fixed for all participants, with Case 3 always preceding Case 4. The objective of Task 2 was to examine the influence of AI advice correctness on diagnostic performance.

To achieve this, we employed a permutation design without replacement. Participants were randomly assigned to one of six unique experimental trajectories, each consisting of two distinct conditions drawn from

three possibilities: Control (no AI), Incorrect AI Advice, and Correct AI Advice ($P(3,2) = 6$). These trajectories covered all ordered pairs of distinct conditions (e.g. Control→Incorrect, Correct→Control).

Because it was not feasible to manipulate the correctness of a real-time LLM output reliably, we constructed a mock-up AI chatbot for the Incorrect and Correct conditions in Task 2. The textual advice

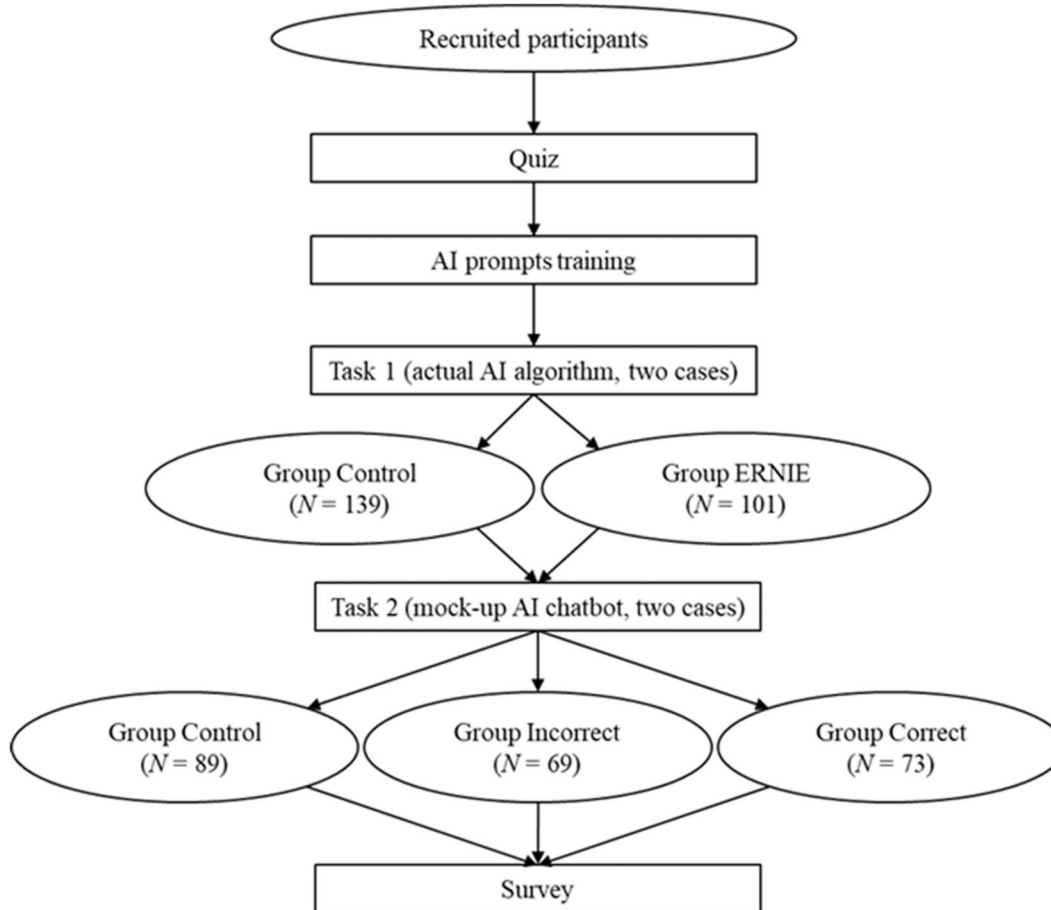


Figure 2. Flowchart of the experiment.

如何给 AI 编写高效的提示词

How to Write Effective Prompts for AI

在与 AI 进行对话时，编写高效的提示词（prompt）至关重要，它能够帮助 AI 更加精确地理解我们的问题，并提供更加合乎期望的回答。以下是一些能帮助您编写高效的提示词的技巧：

一、明确目标 Define Your Goal

在编写提示词前，您应该清楚自己希望从 AI 获取什么信息，完成什么任务。利用关键词来指出您关注的核心内容。

二、提供背景信息 Provide Background Information

补充提供问题相关的背景信息，将关键词与特定的研究领域结合起来，让 AI 更好地定位答案与解释范围。

三、反馈与调整 Feedback and Adjustment

如果 AI 的回答不符合预期，可以通过提供反馈，并调整提示词来逐步引导 AI 朝着理想的方向改善答案。

高效的提示词示例： Example of an Effective Prompt:

张某，女，35 岁。2016 年 6 月 23 日就诊。患者结婚 10 年不孕，经期先后不定，经来腹痛，行而不畅，量少色暗，有小血块，经前乳房胀痛，精神抑郁，烦躁易怒，舌质暗红，苔薄白，脉弦。请从中医的角度，给出辨证判断，证候分析，方剂建议和用药分析。

张某，女，35 岁。患者结婚 10 年不孕，经期先后不定，经来腹痛，行而不畅，量少色暗，有小血块，经前乳房胀痛，精神抑郁，烦躁易怒，舌质暗红，苔薄白，脉弦。请从中医的角度，给出辨证判断、证候分析、方剂建议和用药分析。

问题相关的
背景信息

问题的题干

Figure 3. Sample of AI prompts training.

(including disease judgment, syndrome analysis, and prescription) was pre-designed by the research team. Incorrect advice was initially generated by a real AI and then adjusted to reflect common TCM diagnostic errors. Correct advice was derived from the examination standards. Crucially, the visual interface of the mock-up chatbot was completely consistent with the real AI interface used in Task 1. Because we did not include a direct suspicion probe, we cannot conclusively determine whether participants detected the simulation. However, no participant explicitly questioned the interface, and no significant differences in judgment confidence were observed between the real-AI and mock-up conditions (Appendix Table B4), which is at least consistent with broadly comparable engagement. Pilot testing of the real AI confirmed that it did not verbatim reproduce standard exam answers, suggesting

that its output was based on reasoning rather than simple retrieval.

3.1.2.4. Mid-Task surveys and post-experiment questionnaire. Immediately after completing each of the four cases (across Task 1 and Task 2), participants completed a brief survey measuring their judgment confidence and, for those in AI-exposed conditions, the depth of long logical thinking involved. Upon finishing both tasks, participants proceeded to the fifth and final section: a comprehensive survey that collected data on AI resistance and demographic information.

The entire experiment lasted approximately 30–40 min. Upon completion of the experiment, all participants were fully debriefed. They were informed that a simulated AI system had been used in Task 2 to manipulate advice accuracy. For our primary analysis, we adhered to the Intention-to-Treat (ITT) principle,

Case information

病例

请根据病例摘要，完成中医疾病诊断，并提供治疗方剂及煎服法

疾病诊断:

您的作答会由中医专家进行评估，答案质量将影响实验补贴的发放，请认真作答

Answer box of disease judgment

治疗方剂（请列明克数、煎服法）:

您的作答会由中医专家进行评估，答案质量将影响实验补贴的发放，请认真作答

Answer box of prescription

next page

Figure 4. Settings of answer window.

analysing participants according to their originally assigned groups regardless of actual AI usage. An additional As-Treated analysis (reclassifying non-interactors into the Control group) was conducted as a robustness check to address potential non-compliance (see Appendix Table B2).

3.2. Measures

3.2.1. Diagnostic work performance

Diagnostic work performance was measured by diagnosis accuracy. Two independent raters – a Chief Physician and a Full Professor, both with over 20 years of clinical experience – assessed participant responses

Case information

病例

AI 问答

王某，男，64岁。患者近两年来夜尿次数增多，约24次/夜，近3月无明显诱因开始出现排尿时间延长。昨日饮酒后出现小便次数明显增加，约1小时1次，尿线细，尿后余沥不尽，无肉眼血尿，尿道灼热刺痛，时感小腹灼热。舌红苔黄腻，脉弦数。肛门直肠指检：前列腺增大，表面光滑，质软，有弹性，中央沟消失。腹部B超：前列腺体积增大57mmx48mmx40mm，膀胱残余尿量70mL，余未见异常。

此处文本可以复制、粘贴使用

Suggested prompts

建议提示词

请从中医的角度，给出辨证判断、证候分析、方剂建议和用药分析。

此处文本可以复制、粘贴使用

Outputted AI advice

中央沟消失。腹部B超：前列腺体积增大57mmx48mmx40mm，膀胱残余尿量70mL，余未见异常。请从中医的角度，给出辨证判断、证候分析、方剂建议和用药分析。

***辨证判断：根据患者的临床表现，中医可以辨证为“肾阳亏虚，膀胱气化不利”。

***证候分析：由于患者年龄及病程较长，可能伴有一定的肾阳虚衰表现。患者王某，年事已高，肾阳逐渐亏虚，无力温煦膀胱，导致膀胱气化功能减弱，出现夜尿次数增多、排尿时间延长等症状。肾阳不足，气化不利，水道不畅，故见尿线细、尿后余沥不尽。因此，结合患者的症状，可以辨证为肾阳亏虚，膀胱气化不利。

***方剂建议：针对肾阳亏虚，膀胱气化不利，建议采用温补肾阳，化气行水的方剂，如金匮肾气丸加减治疗。

***用药分析：熟地黄、山药、山茱萸、茯苓、泽泻、牡丹皮、桂枝、附子（炮）（先煎），车前子（包煎），滑石（包煎）。针对患者的具体症状，可根据患者的体质和病情，适当调整药物的剂量和配伍。熟地黄、山药、山茱萸补肾填精，温补肾阳。茯苓、泽泻利水渗湿，助膀胱气化。牡丹皮活血散瘀，助膀胱气化。桂枝、附子温阳化气，助膀胱气化功能恢复。车前子、滑石清热利尿，通利水道，缓解尿道灼热刺痛等症。

Question input box

请输入消息

Figure 5. Settings of assistive AI chatbot.

based on the grading criteria of the National Qualification Examination for Medical Practitioners. To ensure objectivity, both raters were blinded to the experimental conditions (group assignments) during the scoring process. The final score for each participant was calculated as the average of the two raters' scores on a 5-point scale (comprising Disease Judgment and Prescription Analysis; see detailed criteria in Appendix A). Inter-rater reliability was good (Lin's CCC = 0.720), confirming consistent evaluation.

Moreover, the advice provided by the mock-up AI chatbot was also evaluated using the same criteria. The mock-up AI received 4 points for its correct advice and 0 points for its incorrect advice. These AI scores serve as benchmarks for interpreting human performance: if participants in the Group Correct achieve a mean score significantly higher than 4, it suggests that humans can provide additional effort and value beyond accurate AI assistance. Conversely, if participants in Group Incorrect score significantly higher than 0, it indicates that humans can partially remedy AI errors. However, if the Group Incorrect score is significantly lower than the Control Group, it implies that human oversight cannot fully mitigate the negative impact of misleading AI advice.

To better evaluate diagnostic work performance, we recorded the diagnosis time (Variable *task time*) that participants spent to complete the whole diagnostic decision-making process for a single case. It includes the time that used to read the question, interact with AI chatbot and type their answers into the computer. This timer starts when the case page loads and ends when the participant submits their answer (before the survey). Furthermore, we also collected participants' judgment confidence every time they finished a case with the help of AI chatbot through a three-items survey (survey items in Appendix A). Sample question of the survey was 'I am confident about my diagnostic judgment'. Analysing these two variables can help offer a more comprehensive assessment of diagnostic work performance.

3.2.2. Long logical thinking

We asked participants to reflect their depth of long logical thinking during diagnostic decision-making by completing a four-items survey (survey items in Appendix A) after each interaction with AI chatbot. Sample questions of the survey were 'AI assistance makes me think twice' and 'AI assistance makes me consider more about medicine logical reasoning'.

3.2.3. Resistance to AI

A questionnaire was designed to indicate participants' resistance to AI, which was based on the identification

measure by Talke and Heidenreich (2014) and Yang, Ngai, and Wang (2024) (survey items in Appendix A). Sample questions of the survey were 'I worry that using AI is bad for my professional image' and 'I think that AI cannot conduct TCM diagnosis instead of human doctors'. Since individual resistance to AI remains constant (Yang, Ngai, and Wang 2024), we thus placed this questionnaire in the final survey to avoid latent influence of item descriptions on participants' interactions with the AI chatbot in the experiment.

4. Results

4.1. Descriptive statistics and visual comparison

Our final analytical sample consists of 440 observations from 110 participants. We excluded five participants due to data inconsistencies (e.g. conflicting group assignments within cases) and retained only the first complete submission for each participant-case combination to ensure data independence.

The mean diagnosis accuracy score is 1.20 (SD = 0.82) on a 5-point scale. Participants reported a moderately high level of judgment confidence ($M = 3.81$, $SD = 0.85$) and resistance to AI ($M = 3.60$, $SD = 0.46$). Table 1 presents the descriptive statistics and correlation matrix. The magnitude of correlations between key variables is generally low (all $|r| < 0.25$), indicating no significant multicollinearity concerns.

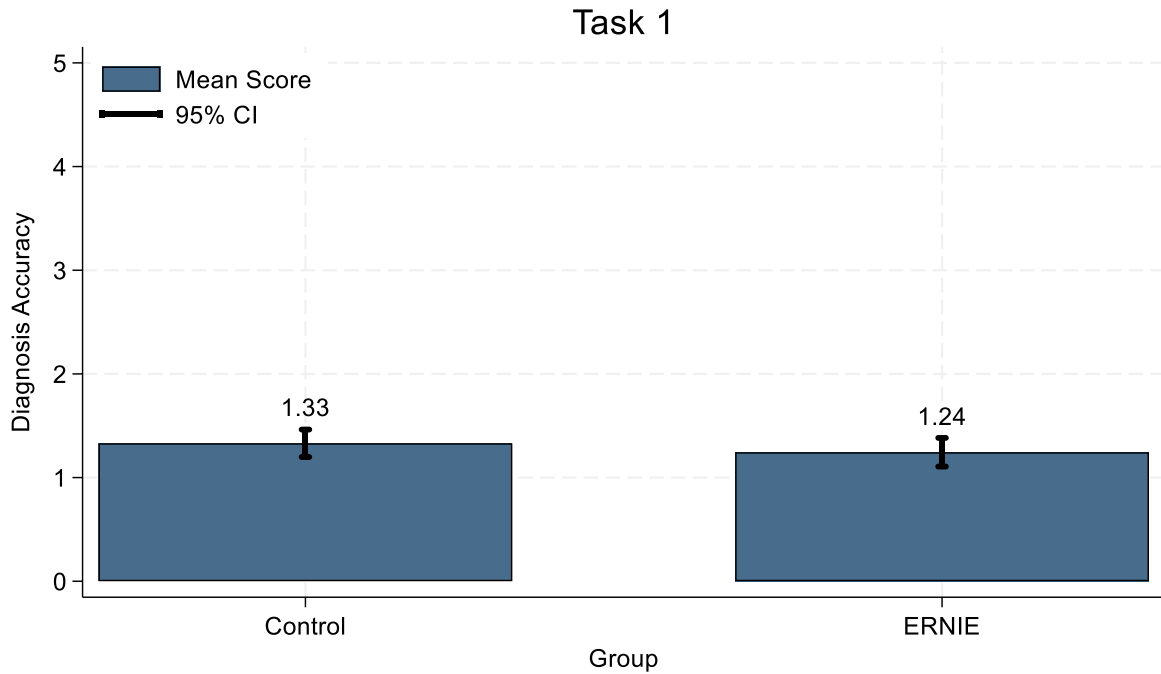
We began by examining the diagnostic accuracy across experimental groups visually to assess the raw effects of AI assistance. Figure 6 presents the mean diagnosis accuracy scores for Task 1 (Control vs. ERNIE), and Figure 7 presents the scores for Task 2 (Control vs. Incorrect vs. Correct). In Task 1, participants in the Control group achieved a mean score of 1.33 (95% CI [1.18, 1.48]), while those in the ERNIE group scored 1.24 (95% CI [1.10, 1.38]). The overlapping confidence intervals suggest no substantial difference between using a general AI assistant and no assistance.

In Task 2, where AI accuracy was manipulated, clear differences emerged. The Control group scored 1.04 (95% CI [0.89, 1.19]). Participants receiving Incorrect AI advice scored substantially lower at 0.63 (95% CI [0.50, 0.76]), with non-overlapping confidence intervals compared to the Control group. Conversely, participants receiving Correct AI advice achieved the highest scores at 1.60 (95% CI [1.45, 1.75]). This visual evidence provides preliminary support for Hypothesis 1, indicating that incorrect AI advice harms performance while correct advice enhances it.

Table 1. Descriptive statistics and correlation matrix.

Variable	Mean	SD	1	2	3	4	5	6	7
1. Diagnosis accuracy	1.20	0.82	1.000						
2. Diagnosis time	375.67	297.61	0.055	1.000					
3. Judgment confidence	3.81	0.85	0.163	0.008	1.000				
4. Resistance to AI	3.60	0.46	0.189	0.037	0.114	1.000			
5. Work Experience	3.98	6.51	−0.005	0.066	0.230	−0.082	1.000		
6. Gender	0.36	0.48	−0.044	0.111	0.105	−0.002	0.470	1.000	
7. Education	1.14	0.37	0.064	−0.064	0.143	−0.128	0.009	0.130	1.000

Notes: N = 440 observations from 110 participants (full sample). Long logical thinking is excluded from this matrix as it was measured only for AI-exposed groups (N = 226, M = 3.85, SD = 0.63). Correlations are based on listwise deletion.

**Figure 6.** Between-group comparison of Task 1.

Notes. Task 1 compares Control vs. ERNIE using a two-sample t-test with unequal variances. Results show no significant difference in Task 1 ($p = 0.377$).

4.2. Main effect of AI accuracy

To test Hypothesis 1 rigorously while controlling for individual heterogeneity and case difficulty, we conducted mixed-effects regressions following the Intention-to-Treat (ITT) principle. Participants were analysed according to their originally assigned groups regardless of actual AI interaction. Standard errors were clustered at the participant level, and case fixed effects (exam question fixed effects) were included.

Table 2 reports the regression results. Model 1 includes only the main effects of AI groups. Model 2 adds Resistance to AI and Task Time. Model 3 includes additional demographic controls. Model 4 employs a high-dimensional fixed-effects estimator absorbing both participant and case fixed effects.

Across all specifications, the results consistently support Hypothesis 1. Compared to the Control group:

Group Incorrect exhibited significantly lower diagnosis accuracy; Group Correct performed significantly better; Group ERNIE showed no significant difference from Control. These results confirm that the accuracy of AI advice causally affects diagnostic performance, with incorrect advice causing significant detriment.

4.3. Moderating roles of human factors

We further examined whether human factors moderate the effect of AI accuracy on diagnostic performance.

4.3.1. Long logical thinking (H2)

Table 3 presents the moderation analysis results about Long Logical Thinking (H2). The measure of long logical thinking was administered after AI interaction; thus, this analysis is restricted to the AI-exposed subsample (N = 272), excluding the Control group where the measure is conceptually inapplicable.

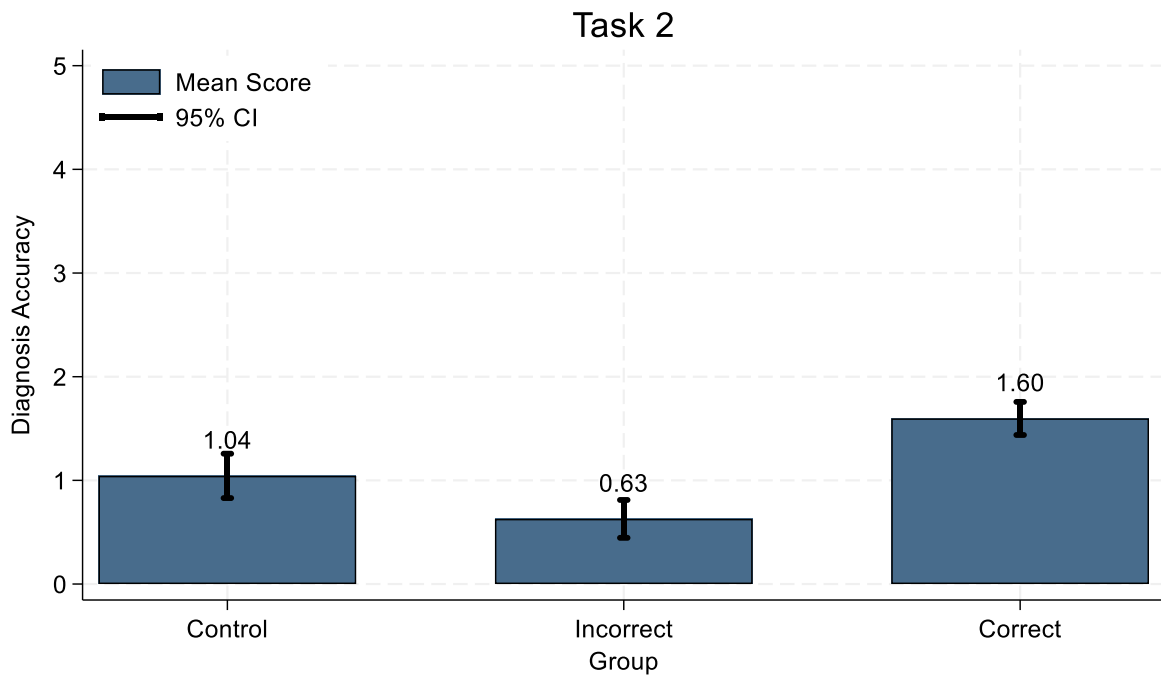


Figure 7. Between-group comparison of Task 2.

Notes. Task 2 compares three conditions (Control, Incorrect, Correct) using pairwise t-tests. The Correct group significantly outperformed both the Control ($p = 0.0001$) and Incorrect ($p < 0.0001$) groups. The Incorrect group also performed significantly worse than the Control group ($p = 0.0046$).

Model 1 in Table 3 shows a significant interaction between AI type and long logical thinking for the Incorrect group ($\beta = -0.342$, $SE = 0.174$, $p = 0.050$). This indicates that participants who engaged in deeper logical thinking performed worse when receiving incorrect AI advice. For the Correct group, the interaction was positive but only marginally significant ($\beta = 0.282$, $p =$

0.086). These findings suggest that long logical thinking does not protect against AI errors but may exacerbate them, possibly due to confirmation bias where users rationalise incorrect AI outputs.

4.3.2. Resistance to AI (H3)

To examine whether resistance to AI moderates the effect of AI assistance, we conducted mixed-effects

Table 2. ITT Analysis (retains participants in originally assigned groups).

DV: diagnosis accuracy	Model 1 (Mixed)	Model 2 (Mixed)	Model 3 (Mixed)	Model 4 (FE)
Group (Base = Control)				
Group ERNIE	-0.086 (0.077)	-0.078 (0.079)	-0.076 (0.079)	-0.083 (0.08)
Group incorrect	-0.443*** (0.121)	-0.447*** (0.123)	-0.447*** (0.122)	-0.457*** (0.13)
Group correct	0.518*** (0.108)	0.524*** (0.107)	0.526*** (0.107)	0.494*** (0.111)
Resistance to AI	-	0.336*** (0.112)	0.365*** (0.109)	-
Task time control	No	Yes	Yes	Yes
Other controls	No	No	Yes	No
Participant FE or RE	Yes (RE)	Yes (RE)	Yes (RE)	Yes
Case FE	Yes	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes	Yes
N	440	440	440	440

Notes: Standard errors in parentheses, p -values in brackets. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Models (1)-(3): Mixed-effects regression with random intercepts for participants. Model (4): High-dimensional fixed-effects regression absorbing participant and case FE. All standard errors clustered at participant level. Other Controls (Model 3) include Education, Gender, and Work Experience. Case fixed effects' refer to exam question fixed effects. Case 1 (question 1) is chosen to be the benchmark. Given the collinearity between case and task, it is unnecessary to include task fixed effects.

Table 3. Moderation by long logical thinking (AI-exposed subsample).

DV: diagnosis accuracy	Model 1 (Mixed)	Model 2 (Mixed)
Group (Base = ERNIE)		
Incorrect	0.45 (0.644)	0.365 (0.658)
Correct	-1.04 (0.672)	-0.954 (0.670)
Long logical thinking	0.20* (0.107)	0.141 (0.112)
Group incorrect $\times$ Long logical thinking	-0.365** (0.172)	-0.342** (0.174)
Group correct $\times$ Long logical thinking	0.305* (0.165)	0.282* (0.164)
Resistance to AI	-	0.26** (0.132)
Task time control	No	Yes
Other controls	No	Yes
Participant RE	Yes	Yes
Case FE	Yes	Yes
Clustered SE	Yes	Yes
Observations	272	272

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. $N = 272$ because Long logical thinking was measured only for AI-exposed groups. Models (1) and (2): Mixed-effects regression with random intercepts for participants.

Table 4. Moderation analysis: the role of resistance to AI on diagnosis accuracy.

DV: diagnosis accuracy	Model 1 (no controls)	Model 2 (base controls)	Model 3 (full controls)
Group (Base = Control)			
Group ERNIE	−0.086 (0.078)	−0.078 (0.079)	−0.076 (0.079)
Group Incorrect	−0.441*** (0.119)	−0.450*** (0.121)	−0.450*** (0.121)
Group Correct	0.529*** (0.106)	0.529*** (0.107)	0.531*** (0.107)
Resistance to AI (Centred)	0.346** (0.134)	0.341** (0.133)	0.369*** (0.132)
Interaction Terms			
ERNIE × Resistance	−0.045 (0.155)	−0.036 (0.154)	−0.034 (0.155)
Incorrect × Resistance	−0.363* (0.201)	−0.366* (0.201)	−0.370* (0.203)
Correct × Resistance	0.279* (0.157)	0.280* (0.158)	0.282* (0.159)
Case FE	Yes	Yes	Yes
Task Time	No	Yes	Yes
Edu/Gender/Experience	No	No	Yes
Participant RE	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes
N	440	440	440
Participants	110	110	110

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Resistance to AI is mean-centred. All models include interaction terms between Group and Centred Resistance. Base Controls = Task Time. Full Controls = Base Controls + Education + Gender + Work Experience.

regressions with interaction terms between AI group and mean-centred resistance to AI (Table 4).

Results show a moderating role of resistance to AI on the effects of both incorrect and correct AI assistance (Incorrect × Resist: $\beta = -0.370$, $p = 0.068$; Correct × Resist: $\beta = 0.282$, $p = 0.076$).

Contrary to our expectation that high resistance would protect against AI errors, participants with higher resistance to AI were actually more negatively affected by incorrect advice. Conversely, high-resistance participants benefited more from correct AI assistance. These results do not support Hypothesis 3. Instead, the interaction signs run opposite to our original prediction, suggesting that resistance to AI may reflect stronger task engagement rather than protective scepticism.

4.3.3. Professional experience (H4)

We also examine whether professional experience moderates the effect of AI assistance (Appendix Table B1). Professional expertise was assessed along two dimensions: professional qualification status (a binary indicator where $qual = 1$ denotes holding a valid medical licence) and work experience Exp, which is the self-reported number of years in clinical practice. Results show that none of the interaction terms of work experience were statistically significant (ERNIE × Exp: $\beta = 0.018$, $p = 0.187$; Incorrect × Exp: $\beta = 0.019$, $p = 0.137$; Correct × Exp: $\beta = -0.017$, $p = 0.269$). These null effects remained consistent across all model

specifications (from no controls to full controls), indicating that work experience alone does not moderate the impact of AI accuracy on diagnostic performance.

We also conducted subsample analyses separating student participants ($qual = 0$, $N = 252$) from licenced doctors ($qual = 1$, $N = 188$). As shown in Appendix Table B3, the detrimental effect of incorrect AI was significantly stronger among students ($\beta = -0.617$, $p < 0.001$) compared to doctors ($\beta = -0.210$, $p = 0.246$). Conversely, the beneficial effect of correct AI was observed in both groups but slightly larger for doctors (students: $\beta = 0.461$, $p < 0.001$; doctors: $\beta = 0.645$, $p < 0.001$).

One possible mechanism is that licenced doctors possess stronger diagnostic schemas and greater confidence in their own clinical judgment, allowing them to resist incorrect AI advice more effectively and leverage correct AI advice more fully, while variation in years of experience alone may be too coarse a measure to capture these oversight capabilities. Taken together, these results provide partial support for Hypothesis 4.

4.4. Robustness checks

We conducted several robustness checks to validate our findings. First, to ensure that our findings are not driven by subjective scoring inconsistencies, we assessed the inter-rater reliability between two independent graders blinded to experimental conditions. For the total diagnosis accuracy score, we observed good absolute agreement (Lin's Concordance Correlation Coefficient [CCC] = 0.720). This agreement was driven by a strong linear correlation (Pearson's $r = 0.756$), indicating consistent ranking of case difficulty across graders, with only minor systematic bias in scoring scale conservatism (bias factor = 0.953). These results confirm that the dependent variable was measured with high reliability, validating the robustness of our main effect estimates against potential measurement noise.

Second, to address concerns regarding the classification of participants who rejected AI assistance (46 observations were assigned to the AI group but had no interaction with the AI), we compared our main Intention-to-Treat (ITT) analysis with an As-Treated analysis where non-interactors were classified into the Control group (see Appendix Table B2). The core findings remained consistent across both grouping approaches: Group Incorrect showed significant negative effects ($\beta = -0.523$, $p < 0.001$ in As-Treated vs. $\beta = -0.443$, $p < 0.001$ in ITT) and Group Correct showed beneficial effects ($\beta = 0.553$, $p < 0.001$ in As-Treated vs. $\beta = 0.518$, $p < 0.001$ in ITT). This confirms that our results are robust to the classification method of non-interacting participants.

Third, we verified the validity of our mock-up AI manipulation. One possible consequence of detecting the simulation would be lower confidence in the mock-up conditions. Appendix Table B4 shows mixed-effects regressions on judgment confidence. Results revealed no significant differences across groups (all $p > 0.10$), suggesting there is no evidence that participants can detect mock-up AI.

5. Discussion and conclusion

5.1. Summary of findings

This study investigates how AI advice accuracy shapes human diagnostic performance and how human factors moderate this relationship. Using a randomised controlled experiment with medical professionals and students, we derive three core findings.

First, AI assistance functions as a double-edged sword: while correct AI advice significantly enhances diagnostic accuracy, incorrect advice causes substantial performance deterioration relative to the control condition. General-purpose AI (ERNIE) yields no significant performance gain, suggesting that unverified AI outputs may offer limited value in high-stakes diagnostic tasks. This detrimental effect of incorrect AI advice on work performance has also been observed in other domains such as personnel selection (Cecil et al. 2024; Schemmer et al. 2022).

Second, contrary to the conventional view that cognitive engagement safeguards against algorithmic errors, we find that long logical thinking marginally exacerbates the detrimental effects of incorrect AI advice. Participants engaging in deeper reasoning performed worse when exposed to errors, likely due to rationalisation mechanisms (Cecil et al. 2024). A similar study of human-AI collaboration by Bilen and Hervé (2025) also identified deficiencies in critical reasoning when individuals interact with chatbots, corroborating our observation that effortful thinking does not guarantee effective error detection.

Similarly, resistance to AI amplifies the magnitude of AI effects in both directions: high-resistance individuals benefit more from correct advice but suffer greater harm from incorrect advice. This indicates that resistance may proxy for task engagement rather than critical immunity.

Finally, domain expertise serves as a robust buffer; licenced physicians are significantly less vulnerable to incorrect AI advice compared to students, whereas years of work experience alone do not confer similar protection.

5.2. Theoretical and practical implications

Theoretically, this study establishes the primacy of AI accuracy over mere presence, demonstrating that AI

acts as a performance multiplier that enhances outcomes when correct but diminishes them below baseline when incorrect. We further identify an ‘effort paradox’, where deeper cognitive engagement and resistance amplify both benefits and harms, suggesting these traits reflect heightened sensitivity to AI quality rather than critical scrutiny, potentially leading to the rationalisation of errors. Additionally, our heterogeneity analysis nuances the role of expertise, revealing that professional qualification buffers against errors while mere tenure does not, thereby distinguishing validated competency from time served.

Practically, these findings necessitate a shift from deployment-focused to accuracy-focused AI integration. Developers must prioritise continuous validation (e.g. Zhang and Zhao 2024; Kong et al. 2023) and uncertainty flagging over general-purpose models to mitigate potential harms. Professional training should evolve from teaching ‘how to use AI’ to ‘how to critique it’, emphasising independent reasoning to counter automation bias. Finally, system designs should encourage critical verification (Isaac et al. 2024), such as requiring preliminary user diagnoses before revealing AI suggestions, thereby anchoring human reasoning and reducing passive acceptance of potentially erroneous outputs.

5.3. Limitations and future research

We note some limitations of our study. First, one of the primary limitations of our research is the sample size. A limited sample can restrict the generalizability of our findings in real-world settings (Hauser, Linos, and Rogers 2017), particularly when it comes to understanding the nuances of AI assistance within a novel context of TCM diagnostic practices. Although our sample offers preliminary insights into the relationship between AI advice and doctor’s diagnostic work performance, a larger, more representative sample would strengthen the robustness of our conclusions. Future research should aim to include a wider range of participants from various industries and across different geographic regions to capture a broader spectrum of experiences and perspectives. This diversification could illuminate how different cultural contexts influence individual attitudes towards AI assistance and application of AI techniques within TCM or other industries, offering a more comprehensive view of AI assistance in work.

Second, our controlled lab design and use of exam-style cases may limit ecological validity. While ensuring standardisation and isolating AI effects, these methods lack the dynamic nature of real patient interactions. We mitigated AI contamination risks by using post-

cutoff cases and human-designed mock-up advice. Future research should validate these findings through field experiments with live clinical cases. Additionally, extending this framework to other legacy sectors (e.g. agriculture, manufacturing) could reveal common strategies for adapting to technological advancements. Finally, future work should address the tension between aspirational AI research roadmaps – which position human oversight and trustworthy AI as foundational principles (Holzinger, Zatloukal, and Müller 2025b) – and the empirically observed human limitations documented in this study.

Third, we acknowledge a limitation regarding the granularity of AI usage data. Due to technical constraints of the experimental platform, we were unable to capture fine-grained interaction logs, such as the exact number of prompts issued or token counts for each participant. This heterogeneity in exposure intensity could potentially influence the observed effects.

Declaration.

We used ChatGPT 5.1 only to improve the language of this paper. Ethics approval was obtained from the IRB committee at Hong Kong University of Science and Technology (Guangzhou). Approval ID: HKUST(GZ)-HSP-2024-0031.

Author contributions

CRedit: **Fei Xiao**: Formal analysis, Methodology, Validation, Writing – review & editing; **Keqing Diao**: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Chenyang Li**: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Yaping Gong**: Conceptualization, Methodology, Project administration, Resources, Supervision, Writing – review & editing; **Beijinni Li**: Data curation, Project administration, Resources, Writing – review & editing; **Jiali Chen**: Data curation, Project administration, Resources, Writing – review & editing; **Xiu-Qiong Fu**: Data curation, Project administration, Resources, Writing – review & editing; **Jian Yan Xie**: Data curation, Project administration, Resources, Writing – review & editing.

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Appendices

Appendix A

Scales of long logical thinking.

No.	Items
1	AI assistance makes me think twice.
2	AI assistance makes me consider more about medicine logical reasoning.
3	AI assistance provides me with new perspective.
4	AI assistance makes me become more cautious.

Notes: All items adopt Likert 5-point scale (1 = strongly disagree, 5 = strongly agree).

Scales of judgment confidence.

No.	Items
1	I am confident about my diagnostic judgment.
2	I think that my diagnostic judgment is correct.
3	I am sure about my diagnostic judgment.

Notes: All items adopt Likert 5-point scale (1 = strongly disagree, 5 = strongly agree).

Scales of resistance to AI (Talke and Heidenreich 2014; Yang, Ngai, and Wang 2024)

No.	Innovation-specific factors	Items
1	Functional barriers	I think that AI advice has value in function. (R)
2		I think that AI advice is good in quality. (R)
3		AI advice has positive impact on my diagnosis. (R)
4		AI advice helps with my diagnosis. (R)
5	Psychological barriers	I worry that using AI is bad for my professional image.
6		I worry that using AI will harm the trust of my patients to me.
7		I think that AI cannot conduct TCM diagnosis instead of human doctors.
8		I think that AI should not be employed in TCM diagnosis.
9		I think that AI can be my work partner. (R)
10		I think that AI can foster the development of TCM. (R)
11		I think that AI can help promote TCM. (R)

Notes: All items adopt Likert 5-point scale (1 = strongly disagree, 5 = strongly agree). Reverse-coded items are marked with (R).

Scoring Rubric (5 Points per Case)

Disease Diagnosis (2 points)

0 pts: Incorrect disease diagnosis.

1 pt: Correct disease diagnosis, but incorrect or missing syndrome differentiation.

2 pts: Correct disease diagnosis and correct syndrome differentiation.

(Other cases: Score at evaluator's discretion)

Prescription Treatment (3 points)

0 pts: Incorrect prescription.

1 pt: Correct prescription name only (no composition/dosage provided).

1–3 pts: Correct prescription with appropriate drug composition and dosage (preparation methods excluded).

(Other cases: Score at evaluator's discretion)

Appendix B

Table B1. Moderation by work experience

DV: diagnosis accuracy	Model 1 (no controls)	Model 2 (base controls)	Model 3 (full controls)
Group (base = control)			
Group ERNIE	−0.086 (0.078)	−0.079 (0.080)	−0.077 (0.080)
Group incorrect	−0.437*** (0.119)	−0.441*** (0.121)	−0.441*** (0.120)
Group correct	0.521*** (0.106)	0.527*** (0.105)	0.529*** (0.106)
Work experience (centred)	−0.006 (0.010)	−0.004 (0.010)	0.001 (0.010)
Interaction terms			
ERNIE × experience	0.019 (0.013)	0.019 (0.014)	0.018 (0.014)
Incorrect × experience	0.020 (0.013)	0.020 (0.013)	0.019 (0.013)
Correct × experience	−0.017 (0.016)	−0.017 (0.016)	−0.017 (0.016)
Case FE	Yes	Yes	Yes
Resistance to AI	No	Yes	Yes
Task time	No	Yes	Yes
Edu/Gender	No	No	Yes
Participant RE	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes
N	440	440	440

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Work Experience is mean-centred. All models include interaction terms between Group and Centred Experience. Base Controls = Resistance to AI + Task Time. Full Controls = Base Controls + Education + Gender.

Table B2. As-treated analysis.

DV: diagnosis accuracy	Model 1 (mixed)	Model 2 (mixed)	Model 3 (mixed)	Model 4 (FE)
Group (base = control)				
Group ERNIE	−0.136 (0.083)	−0.134 (0.085)	−0.135 (0.085)	−0.138 (0.086)
Group incorrect	−0.523*** (0.116)	−0.546*** (0.118)	−0.546*** (0.118)	−0.535*** (0.13)
Group correct	0.553*** (0.11)	0.539*** (0.109)	0.539*** (0.11)	0.506*** (0.115)
Resistance to AI	–	0.323*** (0.111)	0.349*** (0.109)	–
Task time control	No	Yes	Yes	Yes
Other controls	No	No	Yes	No
Participant FE or RE	Yes (RE)	Yes (RE)	Yes (RE)	Yes
Case FE	Yes	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes	Yes
N	440	440	440	440

Notes: We classified participants based on actual AI interaction (i.e. moving non-interactors to the Control group)

Table B3. Subsample analysis by qualification (students vs. doctors).

DV: diagnosis accuracy	Model 1 (qual=0)	Model 2 (qual=0)	Model 3 (qual=1)	Model 4 (qual=1)
Group (base = control)				
Group ERNIE	−0.151 (0.105)	−0.149 (0.104)	0.040 (0.123)	0.042 (0.123)
Group incorrect	−0.617*** (0.156)	−0.616*** (0.156)	−0.210 (0.189)	−0.218 (0.187)
Group correct	0.461*** (0.127)	0.463*** (0.128)	0.645*** (0.170)	0.651*** (0.167)
Resistance to AI	–	0.328** (0.159)	–	0.482*** (0.146)
Task time control	No	Yes	No	Yes
Other controls	No	Yes	No	Yes
Participant FE or RE	Yes (RE)	Yes (RE)	Yes (RE)	Yes (RE)
Case FE	Yes	Yes	Yes	Yes
Clustered SE	Yes	Yes	Yes	Yes
N	252	252	188	188
Participants	63	63	47	47

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Models 1–2: Students (qual = 0). Models 3–4: Doctors (qual = 1).

Table B4. Manipulation check: judgment confidence by AI group.

DV: judgment confidence	Model 1 (Mixed)	Model 2 (Mixed)
Group (base = ERNIE)		
Incorrect	−0.08 (0.106)	−0.089 (0.104)
Correct	−0.091 (0.109)	−0.109 (0.11)
Resistance to AI	–	0.25*
Task time control	–	0.135
Other controls	No	Yes
Participant RE	Yes	Yes
Case FE	Yes	Yes
Clustered SE	Yes	Yes
Observations	272	272